

Comparison across Adaptation Tactics

As a strategy deviates from the initial best execution strategy due to the specification of an adaptation tactic, it results in a new expected cost distribution. The targeted cost and AIM adaptation tactics result in a skewed distribution with a lower expected trading cost but with more “bad” risk. These strategies will take advantage of better market prices by trading at a quicker rate, which results in a lower expected trading cost, but they also trade at a slower rate when prices are less favorable, which increases the chances of incurring costly outliers.

Comparison of the targeted tactic (Equation 8.36) to the AIM tactic (Equation 8.38) shows that the AIM tactic trading rate is $1/3$ of the targeted cost rate. This results in a lower expected cost for the AIM strategy since it allows the fund to participate with favorable price trends for a longer period of time, but since it trades slower it will also expose the order to more market risk which will result in a higher potential for costly outliers when adverse price trends persist.

One strategy that is being used by investors to overcome the “bad” risk issue associated with the targeted cost or AIM adaptation tactic is to set the lower bound or minimum trading rate equal to the original trading rate from step 2—specify trading goal. This then allows investors to take advantage of the better prices when they arise and it will not cause the fund to incur incremental market risk over the constant rate since it will trade no slower than the original trading rate.

The PIM adaptation tactic (Equation 8.40) also results in a skewed distribution, but, unlike the target and AIM tactics, PIM protects investors from “bad” risk by accelerating trading when the adverse momentum begins. The goal of PIM is to complete the order before prices can become too expensive to trade. An advantage of PIM is that it increases “good” risk exposure since it trades slower when there are favorable market prices. But this results in a slightly higher cost on average—but with bad risk protection.

Figure 8.4 provides an example of the difference between the three tactics for a buy order. At the current price the target rate is trading at 30% and both AIM and PIM are at 10%. As prices decline, the target and AIM tactics increase and the PIM tactic decreases. Notice that the target tactic increases three times as quickly (as we determined mathematically). The PIM tactic will decrease down to its minimum rate (which is specified by investor) and in this example is 5%. As prices increase, both the target and AIM tactics decrease and will continue to decrease down to the investor’s specified minimum rate. Since the AIM tactic (10%) is